Turbojet Engine Model

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1 Introduction

A turbojet engine is a propulsion system that generates thrust by accelerating a mass of air through a sequence of thermodynamic and fluid-dynamic processes. The operating principle is based on the Brayton cycle, in which the incoming air is compressed, heated through combustion, and then expanded to produce high-velocity exhaust gases.

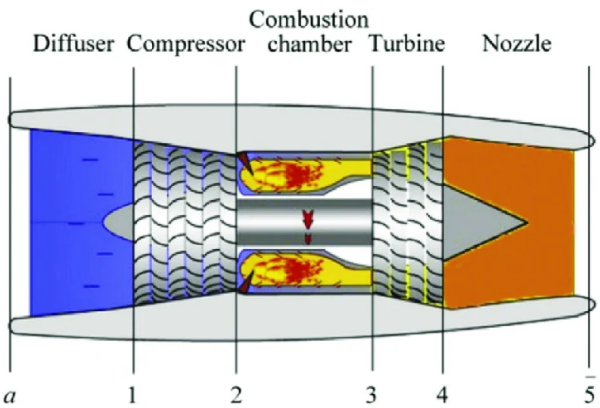

Figure 1: Simplified turbojet engine layout showing the main components and flow path.

The main components of a turbojet engine are:

- **Inlet/Diffuser:** reduces the incoming flow velocity and increases its pressure.
- **Compressor:** increases the total pressure of the air.
- **Combustion chamber:** fuel is injected and burned, raising the total temperature of the flow.

- **Turbine:** extracts part of the energy from the flow to drive the compressor.
- **Nozzle:** expands the flow and converts thermal energy into kinetic energy, producing thrust.

In this project, a simplified turbojet model is developed and implemented in Python. The model is based on a one-dimensional, steady-flow representation of the engine, and includes the main physical processes occurring in each component.

The following assumptions are adopted:

- The working fluid is treated as a perfect gas with constant specific heats.
- The flow is steady and one-dimensional.
- Irreversibilities are taken into account through component efficiencies and pressure losses.
- Total (stagnation) quantities are used throughout the engine core.

The thermodynamic evolution of the flow is modeled through the following steps:

1. Conversion of ambient static conditions into total conditions based on the flight Mach number.
2. Compression process with assigned pressure ratio and efficiency.
3. Energy balance in the combustion chamber to determine the fuel-to-air ratio.
4. Turbine expansion to supply the work required by the compressor.
5. Expansion in the nozzle, including the possible occurrence of choked flow.

In order to analyze the effect of flight conditions on engine performance, the model is evaluated for different subsonic flight Mach numbers ($M_0 < 1$). This allows investigation of how inlet total properties, thrust generation, and specific fuel consumption (SFC) vary with increasing flight velocity.

The performance of the engine is finally assessed in terms of thrust and specific fuel consumption, highlighting the relationship between thermodynamic transformations and propulsion efficiency.

2 Inlet and Diffuser

2.1 Inlet (Freestream to Engine Entry)

The inlet is responsible for converting ambient static conditions into total (stagnation) conditions as a function of the flight Mach number. Due to the incoming velocity of the aircraft, the flow possesses kinetic energy that contributes to the total thermodynamic state.

The total temperature and pressure at the engine inlet are given by:

$$T_{t1} = T_a \left(1 + \frac{\gamma - 1}{2} M_0^2\right) \tag{1}$$

$$P_{t1} = P_a \left(1 + \frac{\gamma - 1}{2} M_0^2\right)^{\frac{\gamma}{\gamma-1}} \tag{2}$$

where M_0 is the flight Mach number and T_a, P_a are ambient static conditions.

2.2 Diffuser

The diffuser reduces the flow velocity before entering the compressor, increasing the static pressure while introducing losses due to irreversibilities.

In this model, the diffuser is assumed to preserve total temperature while accounting for total pressure losses:

$$T_{t2} = T_{t1} \tag{3}$$

$$P_{t2} = \sigma_d \, P_{t1} \tag{4}$$

where σ_d represents the total pressure recovery factor of the diffuser.

3 Compressor

The compressor increases the total pressure of the incoming air, enabling efficient combustion in the subsequent stage. This process requires mechanical work, which is later provided by the turbine.

The compression is characterized by the pressure ratio:

$$\beta = \frac{P_{t3}}{P_{t2}} \tag{5}$$

Under ideal (isentropic) conditions, the process is reversible and adiabatic, therefore:

$$\Delta s = 0 \quad \Rightarrow \quad s_3 = s_2 \tag{6}$$

For a perfect gas, the entropy variation can be expressed as:

$$s_3 - s_2 = c_p \ln\left(\frac{T_{t3s}}{T_{t2}}\right) - R \ln\left(\frac{P_{t3}}{P_{t2}}\right) \tag{7}$$

Imposing the isentropic condition:

$$c_p \ln\left(\frac{T_{t3s}}{T_{t2}}\right) = R \ln\left(\frac{P_{t3}}{P_{t2}}\right) \tag{8}$$

Rearranging the expression:

$$\ln\left(\frac{T_{t3s}}{T_{t2}}\right) = \frac{R}{c_p} \ln(\beta) \tag{9}$$

Using the relation:

$$\frac{R}{c_p} = \frac{\gamma - 1}{\gamma} \tag{10}$$

we obtain the isentropic temperature relation:

$$T_{t3s} = T_{t2}\, \beta^{\frac{\gamma-1}{\gamma}} \tag{11}$$

However, real compressors are affected by irreversibility. This is accounted for through the compressor efficiency, defined as:

$$\eta_c = \frac{T_{t3s} - T_{t2}}{T_{t3} - T_{t2}} \tag{12}$$

Rearranging the expression, the real outlet temperature becomes:

$$T_{t3} = T_{t2} + \frac{T_{t3s} - T_{t2}}{\eta_c} \tag{13}$$

The total pressure at the compressor outlet is, therefore,:

$$P_{t3} = \beta\, P_{t2} \tag{14}$$

4 Combustion Chamber

In the combustion chamber, fuel is injected and burned, causing a significant increase in the total temperature of the flow. The process ideally occurs at nearly constant pressure, although in reality a pressure loss is present due to irreversibilities.

The total pressure at the combustor outlet is modeled as:

$$P_{t4} = \sigma_b \, P_{t3} \tag{15}$$

where σ_b represents the total pressure loss factor in the combustion chamber.

The fuel-to-air ratio is defined as:

$$f = \frac{\dot{m}_f}{\dot{m}_a} \tag{16}$$

The fuel-to-air ratio represents the amount of fuel burned per unit mass of air. It is a key parameter of the model, as it directly determines:

- the increase in total temperature across the combustion chamber,
- the total mass flow entering the turbine,
- the thrust produced by the engine,
- the fuel consumption.

The analysis is based on an energy balance. The chemical energy introduced by the fuel and the enthalpy of the incoming air must equal the enthalpy of the outgoing mixture.

This can be written as:

$$\dot{Q}_{in} + \dot{m}_{in} h_{in} = \dot{m}_{out} h_{out} \tag{17}$$

In this case:

- $\dot{Q}_{in} = \dot{m}_f \, \eta_b \, PCI$
- $\dot{m}_{in} = \dot{m}_a$
- $\dot{m}_{out} = \dot{m}_a + \dot{m}_f$

Substituting into the energy balance:

$$\dot{m}_a c_p T_{t3} + \dot{m}_f \, \eta_b \, PCI = (\dot{m}_a + \dot{m}_f) c_p T_{t4} \tag{18}$$

Dividing by $\dot{m}_a$:

$$c_p T_{t3} + f \, \eta_b \, PCI = (1 + f) c_p T_{t4} \tag{19}$$

Rearranging:

$$f \, \eta_b \, PCI - f c_p T_{t4} = c_p (T_{t4} - T_{t3}) \tag{20}$$

Factoring out f:

$$f \, (\eta_b \, PCI - c_p T_{t4}) = c_p (T_{t4} - T_{t3}) \tag{21}$$

Finally, the fuel-to-air ratio is:

$$f = \frac{c_p (T_{t4} - T_{t3})}{\eta_b \, PCI - c_p T_{t4}} \tag{22}$$

5 Turbine

The turbine extracts energy from the high-temperature gases leaving the combustion chamber in order to drive the compressor through the shaft.

Since fuel has been added in the combustion chamber, the mass flow through the turbine is not only the air mass flow, but the total gas mass flow:

$$\dot{m}_g = \dot{m}_a + \dot{m}_f = (1 + f) \dot{m}_a \tag{23}$$

A power balance is imposed between turbine and compressor. In this model, the work required by the compressor is provided by the turbine through the shaft, accounting for the mechanical efficiency η_m:

$$\dot{W}_c = \eta_m \, \dot{W}_t \tag{24}$$

The compressor power is:

$$\dot{W}_c = \dot{m}_a c_p (T_{t3} - T_{t2}) \tag{25}$$

The turbine power is:

$$\dot{W}_t = \dot{m}_g c_p (T_{t4} - T_{t5}) \tag{26}$$

Substituting into the power balance:

$$\dot{m}_a c_p (T_{t3} - T_{t2}) = \eta_m \, \dot{m}_g c_p (T_{t4} - T_{t5}) \tag{27}$$

Using $\dot{m}_g = (1+f)\dot{m}_a$ and simplifying by $\dot{m}_a c_p$:

$$T_{t3} - T_{t2} = \eta_m (1+f)(T_{t4} - T_{t5}) \tag{28}$$

Rearranging:

$$T_{t5} = T_{t4} - \frac{T_{t3} - T_{t2}}{(1+f)\eta_m} \tag{29}$$

Once the real turbine outlet temperature has been determined, the isentropic outlet temperature is obtained from the turbine efficiency:

$$\eta_t = \frac{T_{t4} - T_{t5}}{T_{t4} - T_{t5s}} \tag{30}$$

Therefore:

$$T_{t5s} = T_{t4} - \frac{T_{t4} - T_{t5}}{\eta_t} \tag{31}$$

Assuming an isentropic expansion between states 4 and 5s, the entropy remains constant:

$$\Delta s = 0 \quad \Rightarrow \quad s_{5s} = s_4 \tag{32}$$

For a perfect gas:

$$s_{5s} - s_4 = c_p \ln\left(\frac{T_{t5s}}{T_{t4}}\right) - R \ln\left(\frac{P_{t5}}{P_{t4}}\right) \tag{33}$$

Imposing the isentropic condition:

$$c_p \ln\left(\frac{T_{t5s}}{T_{t4}}\right) = R \ln\left(\frac{P_{t5}}{P_{t4}}\right) \tag{34}$$

Using the perfect-gas relation, the total pressure at the turbine outlet becomes:

$$P_{t5} = P_{t4} \left(\frac{T_{t5s}}{T_{t4}}\right)^{\frac{\gamma}{\gamma-1}} \tag{35}$$

6 Nozzle

The nozzle expands the flow coming from the turbine and converts thermal energy into kinetic energy, producing the exhaust velocity responsible for thrust generation.

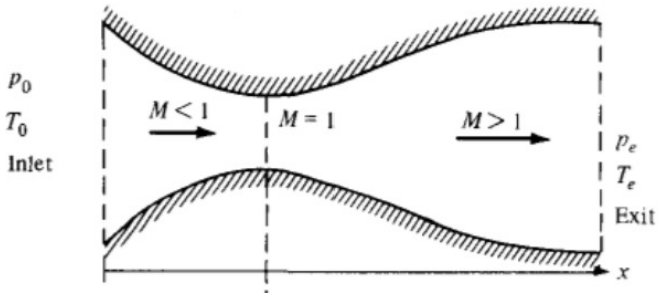

Figure 2: Flow acceleration in a converging nozzle and choked condition at the throat (Mach 1).

The analysis starts from the energy balance of the flow:

$$h_t = h + \frac{V^2}{2} \tag{36}$$

For a perfect gas:

$$c_p T_t = c_p T + \frac{V^2}{2} \tag{37}$$

Rearranging:

$$T_t = T + \frac{V^2}{2c_p} \tag{38}$$

Introducing the Mach number:

$$M = \frac{V}{\sqrt{\gamma R T}} \tag{39}$$

From which:

$$V^2 = M^2 \gamma R T \tag{40}$$

Substituting:

$$T_t = T + \frac{M^2 \gamma R T}{2 c_p} \tag{41}$$

Using the relation:

$$c_p = \frac{\gamma R}{\gamma - 1} \tag{42}$$

we obtain:

$$T_t = T \left(1 + \frac{\gamma - 1}{2} M^2 \right) \tag{43}$$

The corresponding pressure relation is:

$$\frac{P_t}{P} = \left(\frac{T_t}{T} \right)^{\frac{\gamma}{\gamma - 1}} \tag{44}$$

which leads to:

$$P_t = P \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}} \tag{45}$$

6.1 Choked flow condition

The nozzle may become choked, meaning that the flow reaches Mach 1 at the throat and cannot accelerate further in a purely converging geometry.

The critical pressure is defined as:

$$P^* = P_{t5} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}} \tag{46}$$

The choking condition depends on the pressure ratio between the inlet total pressure and the ambient pressure. The choking condition is:

- if $P_a \leq P^*$: choked flow ($M = 1$ at the throat)
- if $P_a > P^*$: not choked

6.2 Exit conditions and velocity

The exit static temperature is obtained assuming isentropic expansion:

$$T_e = T_{t5} \left(\frac{P_e}{P_{t5}} \right)^{\frac{\gamma-1}{\gamma}} \tag{47}$$

The ideal exit velocity is derived from the energy balance:

$$V_{e,id} = \sqrt{2c_p(T_{t5} - T_e)} \tag{48}$$

Accounting for nozzle efficiency:

$$V_e = \sqrt{\eta_n}\, V_{e,id} \tag{49}$$

The exit density is:

$$\rho_e = \frac{P_e}{RT_e} \tag{50}$$

Using the continuity equation, the exit area is:

$$A_e = \frac{\dot{m}_g}{\rho_e V_e} \tag{51}$$

7 Performance

The performance of the turbojet is evaluated in terms of thrust and specific fuel consumption (SFC).

7.1 Thrust

The thrust is obtained from the momentum balance applied to the control volume:

$$F = \dot{m}_g V_e - \dot{m}_a V_0 + (P_e - P_a)A_e \tag{52}$$

where:

- $\dot{m}_g = (1 + f)\dot{m}_a$ is the total mass flow rate of exhaust gases,
- V_e is the exhaust velocity,
- V_0 is the flight velocity,
- $(P_e - P_a)A_e$ is the pressure thrust contribution.

In static conditions ($V_0 = 0$ and $P_e \approx P_a$), the thrust can be approximated as:

$$F \approx (1 + f)\dot{m}_a V_e \tag{53}$$

7.2 Specific Thrust

The specific thrust is defined as:

$$\frac{F}{\dot{m}_a} \tag{54}$$

and represents the thrust produced per unit mass flow rate of air.

7.3 Specific Fuel Consumption

The specific fuel consumption (SFC) is defined as:

$$SFC = \frac{\dot{m}_f}{F} \tag{55}$$

Since:

$$\dot{m}_f = f \cdot \dot{m}_a \tag{56}$$

SFC directly depends on the fuel-to-air ratio and the thrust produced.

A lower SFC indicates a more efficient engine, as less fuel is required to generate the same amount of thrust.

The model allows comparison of thrust and SFC for different flight Mach numbers.

8 Effect of Flight Mach Number

The flight Mach number M_0 affects the inlet total conditions through the conversion of kinetic energy into thermodynamic energy.

Starting from the total energy balance:

$$E_{tot} = E_{int} + E_{cin} \tag{57}$$

For a flowing fluid:

$$c_p T_t = c_p T + \frac{V^2}{2} \tag{58}$$

Rearranging:

$$T_t = T + \frac{V^2}{2c_p} \tag{59}$$

Introducing the Mach number:

$$M = \frac{V}{\sqrt{\gamma RT}} \quad \Rightarrow \quad V^2 = M^2 \gamma RT \tag{60}$$

Substituting:

$$T_t = T + \frac{M^2 \gamma RT}{2c_p} \tag{61}$$

Using the relation:

$$c_p = \frac{\gamma R}{\gamma - 1} \tag{62}$$

we obtain:

$$T_{t1} = T_a \left(1 + \frac{\gamma - 1}{2} M_0^2\right) \tag{63}$$

The corresponding pressure relation is:

$$P_{t1} = P_a \left(1 + \frac{\gamma - 1}{2} M_0^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{64}$$

As M_0 increases, both T_{t1} and P_{t1} increase due to the higher kinetic energy of the incoming flow.

This directly affects the entire thermodynamic cycle:

- Higher T_{t1} leads to higher temperatures throughout the engine.
- Higher P_{t1} results in higher pressure levels after compression.

The flight velocity is:

$$V_0 = M_0 \sqrt{\gamma R T_a} \tag{65}$$

As M_0 increases, the inlet velocity increases and influences the thrust equation:

$$F = \dot{m}_g V_e - \dot{m}_a V_0 + (P_e - P_a) A_e \tag{66}$$

An increase in V_0 tends to reduce the net thrust, since the incoming momentum becomes larger.

At the same time, the variation in inlet total conditions affects the exhaust velocity V_e, and therefore the overall engine performance.

The specific fuel consumption is:

$$SFC = \frac{\dot{m}_f}{F} \tag{67}$$

This demonstrates how propulsion performance is strongly influenced by flight conditions and thermodynamic transformations within the engine. The model highlights the interplay between energy conversion and momentum exchange, providing a clear understanding of how each component contributes to overall engine behavior.